Technical Maths Mark Scheme

Mid Semester Test

Total 80 marks

This teacher mark scheme covers the final 15-question paper. Total marks are allocated to individual assessable steps. Question 7 includes the y-intercept, Question 12 uses a second-derivative sign-change test, and Question 15(c) requires a discriminant argument.

Notice to Markers

M denotes a method mark; **A** an accuracy mark dependent on the associated method; **B** an independent mark. A correct answer may imply M marks unless an asterisk is shown. **M1*** requires the relevant working to be seen.

Accept equivalent exact expressions and valid alternative methods, except where a method is explicitly required. Question 14(b)(ii) requires a final answer to one decimal place; this is part of its final accuracy mark, not an additional mark. No other special significant-figure penalty applies. For requested exact answers, do not replace exact values by rounded decimals.

Error carried forward (ECF): after a numerical error, subsequent valid method marks may be awarded, but accuracy marks are unavailable unless **A1ft** or **B1ft** expressly permits follow-through. Do not award accuracy marks after an unearned associated method mark.

Misreads: apply a single one-mark penalty for an identifiable misread of the question; mark the resulting work consistently, provided the task has not become substantially easier. Miscopying the candidate's own work is an error, not a misread. Disregard crossed-out work.

Question 1–5 **[5 marks]**

Question	1	2	3	4	5
Answer	A	B	B	B	C
Mark	B1	B1	B1	B1	B1

No working is required for these questions.

Question 4 is revised to $x^2 + x - 6 \geq 0$. Since $(x+3)(x-2) \geq 0$, the answer is $x \leq -3$ or $x \geq 2$ (option B).

Question 6 **[5 marks]**

a) [3 marks]

Valid elimination or substitution, e.g. $4p + 6q = 14$, $15p - 6q = 24$. **[M1*]**

$19p = 38 \Rightarrow p = 2$. **[A1]**

$q = 1$. **[A1]**

Marker note: The method must be shown. A correct pair with no supporting working scores 0/3.

b) [2 marks]

Uses $m_1 = -2/3$ and perpendicular gradient $m_2 = 3/2$. **[M1]**

$y + 1 = \frac{3}{2}(x - 4) \Rightarrow \boxed{3x - 2y - 14 = 0}$. **[A1]**

Accept a nonzero integer multiple in the requested form.

Question 7 [7 marks]

a) [2 marks]

Valid factorisation method, e.g. splitting $-11x$ into $-15x + 4x$. **[M1]**

$\boxed{(3x+2)(2x-5)}$. Both marks may be implied by correct factors. **[A1]**

b i) [1 marks]

$\boxed{x = -\frac{2}{3},\ \frac{5}{2}}$; both required. **[B1]**

b ii) [2 marks]

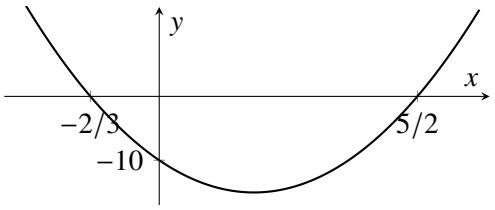
Selects the outside intervals using roots, a sign table or a sketch. **[M1]**

$\boxed{x < -\frac{2}{3} \text{ or } x > \frac{5}{2}}$. Both strict intervals required. **[A1]**

b iii) [2 marks]

Correct upward-opening parabolic shape crossing the x-axis twice. **[B1]**

All intercepts labelled: $(-2/3, 0)$, $(5/2, 0)$ and $\boxed{(0, -10)}$. **[B1ft]**



Marker note: Reference shape only. The vertex is not required. Follow through real roots from (b)(i) for the coordinate mark; the y-intercept must be correct. Shape and coordinate accuracy are marked independently.

Question 8 [7 marks]

a) [2 marks]

$(x-2)^2 + (y+3)^2 = 16$. Centre $\boxed{(2, -3)}$. **[B1]**

Radius $\boxed{4}$. **[B1]**

b i) [2 marks]

$m_{AC} = \dfrac{-k-1}{2k-2}$. **[B1]**

$m_{BC} = \dfrac{7-k}{2k-2}$. **[B1]**

b ii) [3 marks]

Uses $m_{AC} m_{BC} = -1$ since AB is a diameter. **[M1*]**

Forms and solves $5k^2 - 14k - 3 = 0$, or an equivalent quadratic. **[M1]**

$(5k+1)(k-3) = 0 \Rightarrow \boxed{k = 3 \text{ or } k = -\frac{1}{5}}$. **[A1]**

Marker note: Direct substitution into the circle alone scores at most M1 A1 (2/3), since "Hence" requires use of part (b)(i). Neither solution is excluded by the stated conditions.

Question 9 [9 marks]

Revised question: $f(x) = x^3 - 3x^2 + 4$.

a) [1 marks]

Remainder on division by $(x - 1)$ is $f(1) = 1 - 3 + 4 = \boxed{2}$. **[B1]**

b) [3 marks]

Valid division by $(x + 1)$, synthetic division or coefficient comparison. **[M1*]**

Quotient $\boxed{x^2 - 4x + 4}$, remainder zero. **[A1]**

Complete factorisation $\boxed{f(x) = (x + 1)(x - 2)^2}$. **[B1]**

For example, synthetic division gives

-1	1	-3	0	4
		-1	4	-4
	1	-4	4	0

Marker note: The missing x term has coefficient zero. Complete factorisation alone, without evidence of division, scores B1 only.

c) [3 marks]

Differentiates and sets $f'(x) = 3x^2 - 6x = 0$. **[M1]**

Both stationary x values: $x = 0,\ 2$. **[A1]**

Both coordinates $\boxed{(0, 4),\ (2, 0)}$. **[A1]**

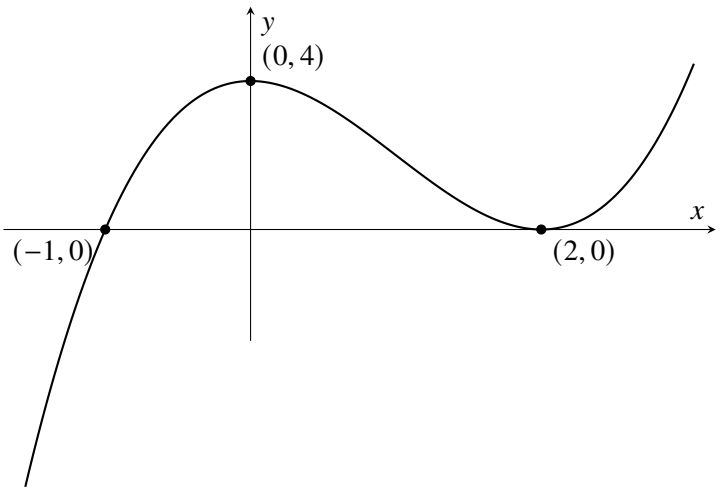
$f''(x) = 6x - 6$ confirms a local maximum at $(0, 4)$ and a local minimum at $(2, 0)$; classification is not separately required in part (c).

d) [2 marks]

Correct positive cubic shape, crossing at $(-1, 0)$ and touching at $(2, 0)$. **[B1]**

Labels $(-1, 0)$, $(0, 4)$ and $(2, 0)$, including both turning points. **[B1ft]**

Marker note: The maximum is also the y-intercept; the minimum is also an x-intercept. Label each coordinate once. Follow through stationary coordinates from (c) for the second mark if consistently plotted; axis intercepts must be correct.



Question 10 [6 marks]

a) [3 marks]

Uses successive differentiation: $y' = 3x^2 - 12x - 12$, $y'' = 6x - 12$. **[M1]**

Marker note: Award the method mark for a valid attempt at both derivatives, allowing a numerical error. Correct roots can imply this unstarred method.

Substitutes into the given equation and solves the resulting quadratic. **[M1]**

$$(6x - 12) + (3x^2 - 12x - 12) = 0 \Rightarrow 3x^2 - 6x - 24 = 0.$$

$(x - 4)(x + 2) = 0 \Rightarrow \boxed{x = -2,\ 4}$. **[A1]**

b) [3 marks]

Integrates correctly, with constant: $y = -\frac{12}{x} + 4x^{3/2} - x + C$. **[M1*]**

Uses $(4, 15)$: $15 = -3 + 32 - 4 + C$, hence determines C. **[M1]**

$\boxed{y = -\dfrac{12}{x} + 4x^{3/2} - x - 10}$, for $x > 0$. **[A1]**

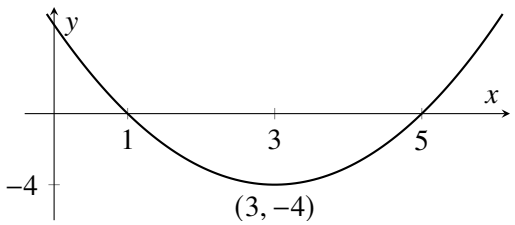
Marker note: Accept $4x\sqrt{x}$ instead of $4x^{3/2}$. A new constant introduced when applying the point is acceptable.

Question 11 [6 marks]

a) [2 marks]

Translation two units to the right. Correct upward parabola with minimum $\boxed{(3, -4)}$. **[B1]**

Both x-intercepts $\boxed{(1, 0),\ (5, 0)}$ labelled. **[B1]**



b i) [1 marks]

$\boxed{g^{-1}(x) = \dfrac{3x + 5}{4}}$. **[B1]**

b ii) [1 marks]

$h'(x) = 3x^2$, so $\boxed{h'(4) = 48}$. **[B1]**

b iii) [2 marks]

Forms $\frac{4(x^3+1)-5}{3} = 85$, or $h(x) = g^{-1}(85) = 65$. **[M1]**

$\boxed{x = 4}$, consistent with $x \geq 0$. **[A1]**

Question 12 [7 marks]

a) [3 marks]

Differentiates and sets $\frac{dy}{dx} = 3x^2 - 12 = 0$. **[M1]**

Both x values: $x = -2,\ 2$. **[A1]**

Both complete coordinates $\boxed{(-2, 21),\ (2, -11)}$. **[A1]**

b) [2 marks]

Uses $\frac{d^2y}{dx^2} = 6x$ and evaluates its sign at both stationary points. **[M1*]**

$y''(-2) = -12 < 0$: local maximum; $y''(2) = 12 > 0$: local minimum. **[A1ft]**

Marker note: Allow follow-through on the stationary x values from (a) if the correct second derivative is used and gives definite classifications.

c) [2 marks]

$y'' = 6x = 0$ gives $x = 0$, and substitution gives $y = 5$. Correct coordinates $\boxed{(0, 5)}$. **[B1]**

$y'' < 0$ for $x < 0$ and $y'' > 0$ for $x > 0$; concavity changes at $x = 0$. **[B1]**

Marker note: A second-derivative sign-change argument is required for the final mark. For example, $y''(-1) = -6 < 0$ and $y''(1) = 6 > 0$, together with the sign change and conclusion, is sufficient. Merely setting $y'' = 0$ is insufficient. Do not award the final mark for a third-derivative argument alone.

Question 13 [7 marks]

a) [2 marks]

Forms $x^2 + 4xh = 300$ and substitutes $h = \frac{300 - x^2}{4x}$ into $V = x^2h$. **[M1*]**

Simplifies correctly to the stated result $\boxed{V = 75x - \frac{1}{4}x^3}$. **[A1]**

Marker note: This is a "Show that" question. The supplied result alone scores no marks.

b) [2 marks]

Differentiates and sets $\frac{dV}{dx} = 75 - \frac{3}{4}x^2 = 0$. **[M1]**

$x^2 = 100$, hence $\boxed{x = 10 \text{ cm}}$, rejecting -10. **[A1]**

c) [3 marks]

Obtains $V'' = -\frac{3}{2}x$ and evaluates at the positive stationary value. **[M1*]**

$V''(10) = -15 < 0$, hence the volume is a maximum. **[A1]**

$V(10) = 750 - 250 = \boxed{500 \text{ cm}^3}$. **[B1]**

Marker note: A valid first-derivative sign-change test also earns the first two marks. The physical domain is $0 < x < 10\sqrt{3}$; stating it is not required.

Question 14 **[9 marks]**

a) [3 marks]

$$\int \left(4x^3 - \frac{6}{x^2} + 2\sqrt{x}\right)\,dx = \boxed{x^4 + \frac{6}{x} + \frac{4}{3}x^{3/2} + C}.$$

x^4. **[B1]**

$+6/x$. **[B1]**

$+\frac{4}{3}x^{3/2}$, with $+C$ in the complete answer. **[B1]**

Marker note: All nonconstant terms correct but C omitted scores 2/3.

b i) [2 marks]

Equates $4x - x^2 = x$ and solves $x(3 - x) = 0$. **[M1]**

Both intersection points $\boxed{(0, 0),\ (3, 3)}$. **[A1]**

b ii) [4 marks]

The curve lies above the line on $0 < x < 3$. Forms $A = \int_0^3 [(4x - x^2) - x]\,dx$. **[M1*]**

Correct antiderivative $\frac{3}{2}x^2 - \frac{1}{3}x^3$. **[A1]**

Substitutes both limits and subtracts in the correct order. **[M1]**

$A = \left[\frac{3}{2}x^2 - \frac{1}{3}x^3\right]_0^3 = \frac{27}{2} - 9 = \frac{9}{2}$. Final positive area $\boxed{4.5 \text{ square units}}$, to one decimal place. **[A1]**

Marker note: Accept separate integrals followed by subtraction, or reverse subtraction followed by an explicit magnitude. A negative final area loses only the final A1. An exact answer $9/2$ alone does not satisfy the stated decimal instruction and loses only the final A1.

Question 15 **[12 marks]**

The paper uses $A(1,2)$, $B(5,2)$ and $C(6,7)$.

a) [4 marks]

Uses the perpendicular bisector of AB: centre has x-coordinate 3. **[M1]**

Write the centre as $(3,t)$. Equal distances from A and C give

$$4+(t-2)^2 = 9+(t-7)^2.$$

Forms a valid equal-radius equation (or equivalent perpendicular bisector). **[M1*]**

Expanding gives $t^2-4t+8 = t^2-14t+58$, so $10t = 50$. Hence $t=5$ and centre $\boxed{(3,5)}$. **[A1]**

$r^2 = 2^2+3^2 = 13$; circle $\boxed{(x-3)^2+(y-5)^2 = 13}$. **[A1]**

Marker note: Accept $x^2+y^2-6x-10y+21=0$. For example, the perpendicular bisector of AC is $x+y=8$; intersecting it with $x=3$ gives the same centre.

b) [2 marks]

Radius gradient $3/2$, so tangent gradient $-2/3$. **[M1]**

$y-2 = -\frac{2}{3}(x-1) \Rightarrow \boxed{l_1 :\ 2x+3y=8}$. **[A1]**

c) [5 marks]

A parallel line has equation $2x+3y=c$, so $y=(c-2x)/3$. Uses this parallel-line form and substitutes it into the circle equation. **[M1*]**

$$9(x-3)^2+(c-2x-15)^2 = 117.$$

Obtains $13x^2+(6-4c)x+(c-15)^2-36=0$. **[A1]**

For tangency sets the discriminant to zero. **[M1*]**

$$(6-4c)^2-52\left[(c-15)^2-36\right] = -36(c-8)(c-34) = 0.$$

Rejects $c=8$ (the given tangent) and obtains $\boxed{l_2 :\ 2x+3y=34}$. **[A1]**

With $c=34$, the intersection equation is $13(x-5)^2=0$. $x=5$ and $y=(34-10)/3=8$, hence $\boxed{Q=(5,8)}$. **[A1]**

Marker note: The second A1 depends on the discriminant method mark. The final A1 also requires the discriminant method and a valid contact-point calculation. Reflection or distance-to-centre alone cannot earn those marks; the parallel-line substitution and correct quadratic may still earn their own marks. Accept an equivalent discriminant calculation in y, or the form $y=-2x/3+b$.

d) [1 marks]

Midpoint of AQ is $(\frac{1+5}{2}, \frac{2+8}{2}) = (3,5)$, the centre; hence AQ is a diameter. **[B1]**

Both endpoints lie on the circle. Accept another complete geometric verification.

Mark Allocation Summary

Question	Part marks	Total
1–5	1 + 1 + 1 + 1 + 1	5
6	3 + 2	5
7	2 + 1 + 2 + 2	7
8	2 + 2 + 3	7
9	1 + 3 + 3 + 2	9
10	3 + 3	6
11	2 + 1 + 1 + 2	6
12	3 + 2 + 2	7
13	2 + 2 + 3	7
14	3 + 2 + 4	9
15	4 + 2 + 5 + 1	12
	Total	**80**

Consistent Marking Standard

Award marks for mathematical achievement, not the number of lines written. Straightforward substitution and rearrangement are not automatically separate marks. In particular, Q11(b)(iii) has one composition method mark and one final accuracy mark; Q14(b)(ii) has no extra mark simply for writing 4.5 instead of $9/2$.

Graph shape and labelled coordinates are marked separately. A sketch need not be to scale. Do not require an unrequested vertex, exact curvature or extra labels. Consistent follow-through on previously calculated roots or turning points is permitted only where stated.

After an earlier numerical error, retain subsequent valid method marks. Do not impose an additional blanket penalty. Accuracy marks still require the correct answer unless marked "ft". Do not repeatedly deduct for the same missing unit or presentational issue; no standalone unit penalty applies unless expressly included in a scoring criterion.

For "Show that", "Hence" and required-method questions, award credit for the specified reasoning. In Q15(c), a reflection-only answer does not demonstrate the required discriminant method. In Q12(c), coordinates can earn B1, but the justification mark requires second-derivative sign change.

Paper Alignment

This scheme corresponds to Technical Math Exam Paper, with $C(6, 7)$ in Q15 and a total of 80 marks. Q7 uses (a), (b)(i), (b)(ii), (b)(iii). Q14(b)(ii) requires one decimal place, so 4.5 is the final answer.

The paper does not explicitly name the required method in Q15(c). This teacher scheme uses the discriminant, as specified for this assessment. To make that requirement clear to students, add "Use the discriminant" to Q15(c). Likewise, Q12(c) is marked using a second-derivative sign-change justification.

Mark Audit

Questions 1–5 contribute five marks. Questions 6–15 contain 75 one-mark scoring lines. Each part subtotal and question total matches those lines. The complete sum is 80.